Section – A

*Attempt **All** Questions* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	What do you mean by EDA? What is the purpose of performing EDA? What is Univariate analysis?	3	1	R	F
2	Identify the level of measurement **(Nominal/Ordinal/Interval/Ratio)** and Justify your reason. (i) The colors of cars in a parking lot are: • Red, Blue, Black, White, Green (ii) The shirt sizes available in a shop are: • Small, Medium, Large, Extra Large (iii) The survey responses for customer satisfaction are recorded as: • Very Unsatisfied, Unsatisfied, Neutral, Satisfied, Very Satisfied (iv) What level of measurement is used to "measure" or "label" someone's religion? (v) ZIP code is an example of which scale of measurement? (vi)Which level of measurement would you use to count the hairs on this man's head?	3	2	U	C
3	Explain the difference between loc and iloc of pandas module with the help of code. How to get all the row indexes and column indexes if df is the dataframe?	3	2	U	C
4	Differentiate between the following: - (i)Population vs Sample (ii)Descriptive Statistics vs Inferential Statistics	3	1	R	F
5	The monthly salaries (in multiple of thousand rupees) of employees are: 25, 28, 32, 35, 37, 38, 40, 42, 45, 60	3	5	A	P

	(i) Calculate the **IQR**. (ii) Using **IQR**, identify any outliers.				

Section – B

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	What do you mean by Data Science? Explain the Data Science process in detail. What are the roles and responsibilities of a Data Scientist?	5	1	R	F
7	Answer the following questions with the help of code snippet only: - a. Create one dictionary of your choice and then create a pandas DataFrame from this dictionary? b. How do you get the first 3 rows and last 2 rows of a DataFrame? c. How do you detect missing values in a DataFrame? d. How do you drop all rows with missing values of a particular column? e. How can you remove duplicate rows from a DataFrame?	5	2	U	C
8	What is the output of the following code snippets? import numpy as np (i) a = np.arange(5) print(a[::-1]) (ii) a = np.array([[1, 2], [3, 4], [5, 6]]) print(a[1, 0]) (iii) a = np.arange(9).reshape(3, 3) print(a.T) (iv) a = np.arange(12).reshape(3, 4) print(a[::2, 1::2]) (v) a = np.array([[1, 2], [3, 4]]) print(np.sum(a, axis=0))	5	5	A	P

	(vi) a = np.array([[1, 2, 3], [4, 5, 6]]) print(a.ndim) (vii) a = np.arange(12).reshape(3, 4) print(a[:, 1]) (viii) a = np.array([1, 2, 3]) b = np.array([4, 5, 6]) print(np.dot(a, b)) (ix) a = np.array([1, 2, 3, 4, 5]) print(np.std(a)) (x) a = np.array([1, 2, 3, 4]) b = np.array([10, 20, 30, 40]) print(np.concatenate((a, b))				